INCH-POUND

MIL-A-9286C (USAF)
AMENDMENT 2
26 AUGUST 1994
SUPERSEDING
AMENDMENT 1
29 JANUARY 1975

MILITARY SPECIFICATION
AF AMPLIFIER AM-476( )/AIC-10

This amendment forms a part of Military Specification MIL-A-9286C (USAF) dated 27 May 1969 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1: Delete: "MIL-E-5272" and its title.

* Delete: The title of MIL-STD-461 and substitute: "Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility".

* Delete: The title of MIL-STD-704 and substitute: "Aircraft Electric Power Characteristics".

* Delete: The title of MIL-STD-781 and substitute: "Reliability Testing for Engineering Development, Qualification, and Production".

* Add: "MIL-STD-810, Environmental Test Methods and Engineering Guidelines".

* Add: The headings "HANDBOOK" and "MILITARY" and list "MIL-HDBK-781, Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production"

PAGE 2

* 3.2.1.2: Delete in its entirety and add:

"Amplifier Design. This amplifier is to be designed to meet the specifications of MIL-A-9286C. The design of the amplifier module and the choice of components used in the amplifier are up to the discretion of the contractor. Drawings 58C12600 and 53C13654 are not applicable to this specification for use as schematics for manufacturers to duplicate, but are included because they contain the pin layouts and physical dimensions of the amplifier and locations of connectors and mounting hardware."

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* Add: "3.2.1.2.1 Electrical Connectors. Connectors J101 and J102 are NSN 5935-00-201-2859. Part numbers and manufacturers are: P/N 458430-5, General Electric Co. Camden, NJ 08102 and P/N 126-270 Amphenol Corp. Wallingford, CT 06492."

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* 3.3: Delete in its entirety and add:

"Environmental Testing. Environmental testing shall be conducted as discussed in paragraph 4.5.3."

PAGE 4

Table II: Change the frequency response at 3000 Hz to Plus 1.0 minus 2.0 dB.

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3.6.2.5: Add: "The AGC attack time shall be critically damped."

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* Add: "3.6.3.1 Talk-Listen Test. Two amplifiers shall be tested as described in paragraph 4.5.2.4."

* Add: "3.6.5 Load-Simulator Continuity Test. This test shall be conducted per paragraph 4.5.2.6."

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* Table VI: Add under the following headings:

INSPECTION OR TEST	TEST PROCEDURE OR REQUIREMENT	PREPRODUCTION	INDIVIDUAL	SAMPLING
Headset Amplifier				
"AGC Disabling	3.6.1.3	3		all"

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Table VI: Add: "All" under column heading INDIVIDUAL for AGC Attack Time.

* Table VI: Delete the following Environmental Tests:
  Fungus
  Salt Spray
  Temperature Shock
and add under Environmental Tests (to be performed on 2 PREPRODUCTION amplifiers only):
  Salt Fog
  Explosive Atmosphere

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* Table VI: Add under the heading Microphone Amplifier:

INSPECTION OR TEST	TEST PROCEDURE OR REQUIREMENT	PREPRODUCTION	INDIVIDUAL	SAMPLING
"Location of AGC Breakpoint	4.5.2.2.6.1	all		all"

* Table VI: Add the following heading and information at the end of the table:

INSPECTION OR TEST	TEST PROCEDURE OR REQUIREMENT	PREPRODUCTION	INDIVIDUAL	SAMPLING
"Environmental Stress Screening	4.5.4	all	all	all"

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* 4.5.1: Insert " weight," after "materials," in line 2.

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4.5.2.2.5: Substitute: "sustain a value" for "rise to" and add "The AGC shall be checked for critical damping using a storage scope."

4.5.2.2.5: Line 2: Change "input of 1.3 volts (rms)" to "at input levels of 0.4, 1.3, and 4.2 volts (rms)".

* Add: "4.5.2.2.6.1 Location of AGC Breakpoint. The AGC breakpoint shall be located and must occur in the input voltage range specified in paragraph 3.6.2.5.1."

* Add: "4.5.2.2.8 Output-to-Input Feedback. This test shall be conducted per paragraph 3.6.2.5.3."

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* 4.5.2.5.1: Delete in its entirety and add:

"Reliability Test. The preproduction samples of the equipment shall be tested by the contractor for reliability in accordance with MIL-STD-781 and MIL-HDBK-781 using test plan V-D and the parameters θ_0 = 7500 and θ_1 = 2500. The accept-reject criteria shall be in accordance with test plan V-D of MIL-HDBK-781. The performance parameters to be monitored during the reliability testing shall be as specified in the individual tests of Table VI of this document. These parameters shall be included in the detailed reliability test procedure prepared by the supplier."

* 4.5.2.5.2 through 4.5.2.5.6: Delete text.

* Add: 4.5.2.6.1 High Voltage Load Simulator. This test shall be conducted per paragraph 3.6.4."

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* 4.5.3: Delete in its entirety and add:

"Environmental Tests. The applicable environmental tests shall be performed in accordance with MIL-STD-810, for the specified class of the component under test. Tests not listed do not waive the requirements in designing the component for airborne use or storage under extreme conditions. During shock, vibration, and temperature tests the amplifier shall be mounted in an Interphone Control C-823A/AIC-10. The following tests shall be conducted on the components of intercommunication set:

a. Temperature-Altitude, Method 504, Procedure I
b. Humidity, Method 507, Procedure I
c. Salt Fog, Method 509, Procedure I
d. Explosive Atmosphere, Method 511, Procedure I
e. Vibration, Method 514, Procedure I, Curve M, Parts 1, 2, and 3
f. Shock, Method 516, Procedures I and III

* Table VII: Delete table in its entirety.

* Add: 4.5.4 Environmental Stress Screening (ESS). Each amplifier shall be non-energized and subjected to 25 temperature cycles from -54°C to +71°C chamber temperature at a temperature rate change of 15°C per minute minimum Temperature dwell at temperature extremes shall be that time required for the item to stabilize in temperature. Stabilization occurs when the temperature of the part of the amplifier having the longest thermal lag has met or exceeded the maximum or minimum temperature for 5 minutes."

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* 6.1: Insert "Intended" in front of the word "Use" in the title of the paragraph.

* 6.2: Add at the end of the paragraph: "The contractor has the responsibility to obtain all Military Specifications and Standards.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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